Branch Cash Flow Insights

Optimizing liquidity, reducing shortages, and enabling data-driven branch operations

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The Problem

Cash Shortages

Uneven distribution across branches, recurring stockouts

Liquidity Risk

High withdrawal volumes signal pressure at key branches

Idle Cash

Excess cash sitting unused while others face shortfalls

Forecasting Gap

Monthly deposit fluctuations make planning difficult

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Project Overview

Analyzing daily deposits, withdrawals, inter-branch transfers, and ATM transactions to uncover cash utilization patterns, branch performance disparities, and operational risks — enabling smarter cash allocation.

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Data & Methodology

Dataset

- 3,000+ rows, 10+ columns
- Branch ID, Region, Deposit & Withdrawal Amounts
- Opening/Closing Balance, Transaction Count
- Cash Shortage Flag

Python EDA

- Data cleaning with **pandas**
- Feature engineering: Net Cash Flow, Cash Difference
- Visualization: **seaborn** & **matplotlib**
- Anomaly detection + SQL integration

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Key SQL Findings

838M

Total Deposits

739M

Total Withdrawals

99M

Net Cash Flow

129

Cash Shortage Branches

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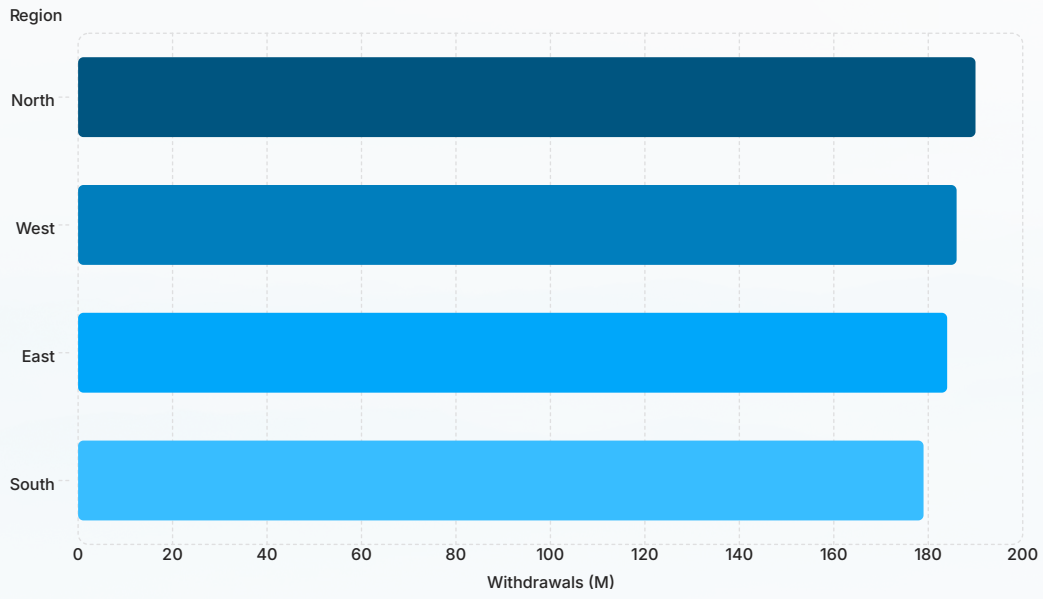
Regional Cash Distribution



Regional Deposits

- **North:** 214.5M
- **West:** 214.5M
- **East:** 207.3M
- **South:** 202.1M

Avg Balance by Region

- South & West: ~540K
- North: ~534K
- East: ~527K

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Monthly Withdrawal Trend

Withdrawals fluctuate between **55.7M (Nov)** and **69.8M (Mar)** — March peaks signal seasonal liquidity pressure.

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Key Insights

🔴 High-Risk North Region

~190M withdrawals — highest demand zone

⚠️ Top Branches at Risk

B29 (18.7M), B35 (18.4M), B8 (18.4M) — peak withdrawal branches

💧 4.3% Shortage Rate

129 branches flagged — active liquidity gaps

📈 Volatile Deposits

65M–77M range monthly — inconsistent inflow patterns

💤 Idle Cash Risk

High closing balances in select branches — capital inefficiency

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Business Recommendations

Focus on High-Risk Branches

Prioritize B29, B35, B8 for proactive cash replenishment

Regional Strategy

Reallocate from low-demand to high-demand zones

Optimize Cash Allocation

Reduce idle cash, lower holding costs

Improve Forecasting

Data-driven liquidity planning for better operational efficiency

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